



Post Office Box 2000, Decatur, Alabama 35609-2000

January 29, 2026

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Browns Ferry Nuclear Plant, Unit 1
Renewed Facility Operating License No. DPR-33
NRC Docket No. 50-259

Subject: **Licensee Event Report 50-259/2025-001-01**

Reference: **Letter from TVA to NRC, "Licensee Event Report 50-259/2025-001-00," dated September 30, 2025 (ML25273A039)**

The enclosed Licensee Event Report provides details of a SCRAM due to low reactor water level on Browns Ferry Nuclear Plant, Unit 1. The Tennessee Valley Authority is submitting this report in accordance with Title 10 of the Code of Federal Regulations (10 CFR) 50.73(a)(2)(iv)(A), as any event or condition that resulted in a manual or automatic actuation of the Reactor Protection System and Primary Containment Isolation System. Revision 1 to this report includes information from the cause evaluation that was not previously reported.

There are no new regulatory commitments contained in this letter. Should you have any questions concerning this submittal, please contact David J. Renn, Site Regulatory Compliance Manager, at (256) 729-2636.

Respectfully,

A handwritten signature in blue ink, appearing to read "D. Komm", is written over a horizontal line.

Daniel A. Komm
Site Vice President

Enclosure: Licensee Event Report 50-259/2025-001-01 – Reactor SCRAM due to Low Reactor Water Level

cc (w/ Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Browns Ferry Nuclear Plant
NRC Project Manager - Browns Ferry Nuclear Plant

ENCLOSURE

**Browns Ferry Nuclear Plant
Unit 1**

Licensee Event Report 50-259/2025-001-01

Reactor SCRAM due to Low Reactor Water Level

See Enclosed

NRC FORM 366 (04-02-2024)		U.S. NUCLEAR REGULATORY COMMISSION		APPROVED BY OMB: NO. 3150-0104		EXPIRES: 04/30/2027						
		LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)				Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov , and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk ail: pira_submission@omb.eop.gov . The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.						
1. Facility Name Browns Ferry Nuclear Plant, Unit 1				☒ 050 ☐ 052		52. Docket Number 00259						
						3. Page 1 OF 6						
4. Title Reactor SCRAM due to Low Reactor Water Level												
5. Event Date			6. LER Number			7. Report Date		8. Other Facilities Involved				
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number		
08	01	2025	2025	- 001 -	01	01	29	2026	N/A	05000XXX		
									Facility Name	Docket Number		
									N/A	05000XXX		
9. Operating Mode 1						10. Power Level 100						
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)												
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)		
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)		
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)		
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☒ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)		
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)		
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)		☐ 73.1200(f)		
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)		
☐ 20.2203(a)(2)(iv)				☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)		
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)						
☐ OTHER (Specify here, in abstract, or NRC 366A).												
12. Licensee Contact for this LER												
Licensee Contact Justin Garner, Site Compliance Engineer								Phone Number (Include area code) 256-729-7955				
13. Complete One Line for each Component Failure Described in this Report												
Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS			
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
4. Supplemental Report Expected)					15. Expected Submission Date					Month	Day	Year
☒ No					☐ Yes (If yes, complete 15. Expected Submission Date					N/A	N/A	N/A
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)												
On August 1, 2025, at 2350 Central Daylight Time, Browns Ferry Unit 1 experienced an automatic scram due to low reactor water level. A low reactor water level of +2 inches resulted in a valid actuation of the Reactor Protection System which caused all of the rods to insert. During the scram response, there was a valid actuation of the Primary Containment Isolation System groups 2, 3, 6, & 8. Upon receipt of these signals, all components actuated as required. Following the scram, reactor water level lowered below the -45 inch setpoint, actuating High Pressure Coolant Injection and Reactor Core Isolation Coolant and tripped both Reactor Recirculation pumps as required. Operations responded and stabilized the plant. The root cause was determined to be the installed Foxboro Feedwater Level Control system design and subsequent reviews failed to identify the single point vulnerability and inherit potential failure modes for the Foxboro System. The root cause will be addressed through elimination of the single point vulnerability in the Reactor Feed Pump speed control system. The Tennessee Valley Authority is submitting this report in accordance with Title 10 of the Code of Federal Regulations 50.73(a)(2)(iv)(A), as any event or condition that resulted in manual or automatic actuation of the Reactor Protection System and the Primary Containment Isolation System.												

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME Browns Ferry Nuclear Plant, Unit 1	☒ 050	2. DOCKET NUMBER 00259	3. LER NUMBER		
	☐ 052		YEAR 2025	SEQUENTIAL NUMBER - 001	REV NO. - 01

NARRATIVE**I. Plant Operating Conditions before the Event**

At the time of discovery, Browns Ferry Nuclear Plant (BFN), Unit 1, was in Mode 1 at 100 percent power. There were no impacts on Units 2 and 3.

II. Description of Event**A. Event Summary**

On August 1, 2025, at 2350 Central Daylight Time (CDT), Browns Ferry Unit 1 experienced an automatic scram due to low reactor water level. A low reactor water level of +2 inches resulted in a valid actuation of the Reactor Protection System (RPS)[JC] which caused all of the rods to insert. During the scram response, there was a valid actuation of the Primary Containment Isolation System (PCIS)[JM] groups 2, 3, 6, & 8. Upon receipt of these signals, all components actuated as required. Following the scram, reactor water level lowered below the -45 inch setpoint, actuating High Pressure Coolant Injection (HPCI)[BJ] and Reactor Core Isolation Coolant (RCIC)[BN] and tripped both Reactor Recirculation pumps as required. Operations responded and stabilized the plant. Event Notification 57847 was made to the NRC Operations Center on August 2, 2025, at 0429 EDT.

The Tennessee Valley Authority (TVA) is submitting this report in accordance with Title 10 of the Code of Federal Regulations (10 CFR) 50.73(a)(2)(iv)(A), as any event or condition that resulted in manual or automatic actuation of the RPS and the PCIS.

B. Status of structures, components, or systems that were inoperable at the start of the event and that contributed to the event

There were no structures, systems, or components (SSCs) whose inoperability contributed to this event.

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1. FACILITY NAME

Browns Ferry Nuclear Plant, Unit 1



050



052

2. DOCKET NUMBER

00259

3. LER NUMBER

YEAR

2025

SEQUENTIAL
NUMBER

- 001

REV
NO.

- 01

NARRATIVE**C. Dates and approximate times of occurrences**

<u>DATE AND APPROXIMATE TIMES</u>	<u>OCCURRENCE</u>
August 1, 2025, at 23:48:48.020	Foxboro Field Bus Module (FBM) BFN-1-XM-046-0097/52, Reactor Feedwater Level Control System Reactor Feed Pump speed demand, failed
August 1, 2025, at 23:50:13.030	BFN-1-XM-046-0097/52 Reinitialized
August 1, 2025, at 23:50:14.920	RFWLCS PID output to Woodward Governor to 0 (zero)
August 1, 2025, at 23:50:24.450	Unit 1 SCRAM (Event Notification report 57847 was reported to the NRC for this event.)

D. Manufacturer and model number of each component that failed during the event

The manufacturer and model number the component that failed was the Foxboro FBM204, Reactor Feedwater Level Control System (RFWLCS) Reactor Feed Pump speed demand, Part number P0914SY 0H (BFN-1-XM-046-0097/52).

E. Other systems or secondary functions affected

No other systems or secondary functions were affected.

F. Method of discovery of each component or system failure or procedural error

The momentary reset of the Reactor Feedwater Pump speed demand to minimum speed resulted in an automatic Unit 1 SCRAM on low reactor water level.

G. The failure mode, mechanism, and effect of each failed component

The Unit 1 RFWLCS speed demand for all three Reactor Feed Pumps failed momentarily, resulting in an automatic Reactor SCRAM on low reactor water level. The analog output Fieldbus module launched a re-initialization process during normal plant operation and in accordance with the system programming lowered the RFP speed demand to 600 RPM.



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	☐ 052		YEAR 2025	SEQUENTIAL NUMBER - 001	REV NO. - 01

NARRATIVE

H. Operator actions

Operations responded and stabilized the plant following the reactor scram. Reactor water level was maintained via the condensate system. Decay heat was removed by bypass valves to the main condenser.

I. Automatically and manually initiated safety system responses

HPCI and RCIC automatically initiated at -45" reactor water level, as designed.

III. Cause of the event

A. Cause of each component or system failure or personnel error

The root cause was determined to be the installed FWLCS design and subsequent reviews failed to identify the single point vulnerability and inherit potential failure modes for the Foxboro system.

B. Cause(s) and circumstances for each human performance related root cause

None. Analysis determined that there was no active error and that any potential human performance issues occurred more than twenty years prior.

IV. Analysis of the event

Review of Plant Engineering Data Satellite System showed that Reactor Feed Pump Turbine (RFPT) A speed (DFW061), RFPT C speed (DFW063), and LCV-3-53 B signal (DFW065) failed all on the same FBM (1FL502) at 23:48:48.020 (immediately prior to the event) and recovered at 23:50:13.030. The scenario was reproduced on the FWLCS test bed and supports the cause determination.

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NARRATIVE**V. Assessment of Safety Consequences**

The plant responded as designed, while maintaining defense-in-depth for nuclear safety. All nuclear safety systems functioned as designed. At no time was the health and safety of the public at risk.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

No other systems or components could have performed the same function, and the failure was determined to be an SPV. Following a reboot, the FWLCS reset and functioned as designed.

B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

This event did not occur when the reactor was shut down.

C. For failure that rendered a train of a safety system inoperable, estimate of the elapsed time from discovery of the failure until the train was returned to service

This event did not render a train of a safety system inoperable.

VI. Corrective Actions

Corrective Actions are being managed by the TVA's Corrective Action Program under Condition Report (CR) 2029788.

A. Immediate Corrective Actions

The scenario was reproduced on the FWLCS test bed confirming the cause. The failed component was replaced.

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	☐ 052		<table border="1"><tr><th>YEAR</th><th>SEQUENTIAL NUMBER</th><th>REV NO.</th></tr><tr><td>2025</td><td>- 001</td><td>- 01</td></tr></table>	YEAR	SEQUENTIAL NUMBER	REV NO.	2025
YEAR	SEQUENTIAL NUMBER	REV NO.					
2025	- 001	- 01					

NARRATIVE**B. Corrective Actions to Prevent Recurrence or to reduce the probability of similar events occurring in the future**

The Root Cause will be addressed through Corrective Action to Prevent Recurrence Action 2029788-002, which will eliminate the SPV in the Unit 1 Foxboro FWLCS. Additionally, Corrective Action 2029788-003 will update the Equipment Reliability classification for component BFN-1-XM-046-0097/52 based on the discovery of the SPV. Corrective Action 2029788-004 will verify that Foxboro system vulnerabilities are included in the distributed control system hazard review per CR 2029991.

VII. Previous Similar Events at the Same Site

There were no previous similar events at BFN within the last 5 years.

VIII. Additional Information

There is no additional information.

IX. Commitments

There are no new commitments.

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX].